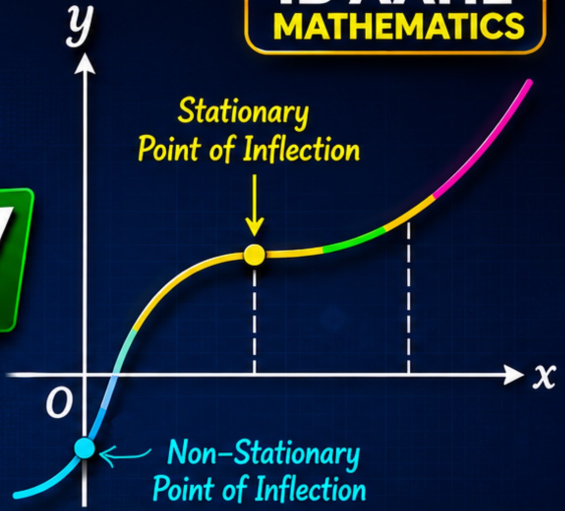
HOW TO FIND STATIONARY & NON-STATIONARY POINTS OF INFLECTION IN 4 EASY STEPS

FOR IB AAHL

IB AAHL MATHEMATICS



- Find y' and y''
- Find Critical Points
- Check Change in y''
- Identify Points of Inflection

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9326363793

Question: Find stationary and non-stationary points of inflection of the following function:

$$f(x) = \frac{1}{4}x^4 - x^3$$

Step (1) : Differentiate the function:

$$f'(x) = \frac{1}{4}\left(4x^3\right) - 3x^2$$

$$f'(x) = x^3 - 3x^2$$

and $$f''(x) = 3x^2 - 6x$$

Step (2): Set $f''(x) = 0$ and find values of x.

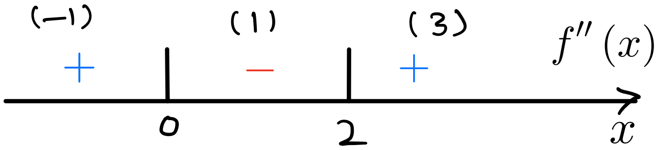
$$f''(x) = 3x^2 - 6x$$

$$3x^2 - 6x = 0$$

$$3x(x - 2) = 0$$

$$x = 0, x = 2$$

Step (3): Test for sign change(Sign diagram).



$$f''(x) = 3x(x-2)$$

$$f''(-1) = 3(-1)(-1-2) = 9(+)$$

$$f''(1) = 3(1)(1-2) = -3(-)$$

$$f''(3) = 3(3)(3-2) = 9(+)$$

Sign of $f''(x)$ changes from (positive) + to (negative)– around x = 0

Hence x = 0 is point of inflection

And sign of $f''(x)$ changes from (negative) – to (positive)+ around x = 2

Hence x = 2 is point of inflection

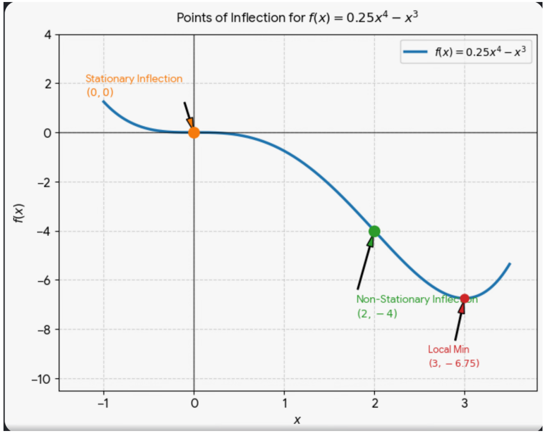
Step (4): Classify using $f'(x)$:

$$f'(x) = x^3 - 3x^2$$
$$= x^2(x-3)$$

at $x = 0$

$$f'(0) = 0^2(0-3) = 0$$

since gradient zero. Therefore, (0, 0) is stationary point of inflection.



And

$$f'(2) = (2)^2(2-3) = -4 \neq 0$$

As gradient is not zero. Therefore, (2, −4) is non–stationary point of inflection.

$$f(x) = \frac{1}{4}x^4 - x^3$$
$$f(0) = \frac{1}{4}(0)^4 - (0)^3$$
$$f(2) = \frac{1}{4}(-2)^4 - (2)^3 = -4$$

End of Lesson

Contact me on: 9326363793

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